

KRRISH BHASIN

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[GitHub](#) | [LinkedIn](#)

EDUCATION

BITS Pilani, K.K. Birla Goa Campus Goa, India | Expected May 2028 B.E. (Hons.) Computer Science

Relevant coursework: Linear Algebra, Probability & Statistics, Foundations of Data Science, Machine Learning, OOP

Achievement: Codeforces Specialist (peak rating 1546)

EXPERIENCE

ML Research Intern, Indian Institute of Technology Bombay Mumbai | Aug 2025 – Feb 2026 Natural Language Processing / Machine Translation Lab, advised by Prof. Ganesh Ramakrishnan

- Researched neural machine translation for Indic languages with transformer models (IndicTrans2) across Hindi–English and multi-Indic pairs.
- Fine-tuned models in Fairseq on large parallel corpora; benchmarked quality with BLEU and chrF across language pairs. – Built an LLM post-correction pipeline with domain-specific glossary injection, improving translation consistency for technical and rare-domain vocabulary.
- Contributed to the Udaan OCR/MT pipeline for digitizing Indian-language educational content; diagnosed tokenization mismatches and stabilized training runs.
- Ran controlled ablations comparing fine-tuned vs. zero-shot performance and presented findings to the lab.

Software Engineering Intern, Hodo Stays Bengaluru | May 2025 – Jul 2025 LLM-powered guest communication for a hospitality SaaS platform

- Designed and shipped a multi-channel (WhatsApp, email, in-app) guest messaging system that automated routine guest communication.
- Engineered a three-class intent classifier (booking / maintenance / general) reaching **~91% accuracy** on an internal test set, cutting manual triage.
- Designed prompt templates with dynamic context injection (guest name, property, booking status) to produce brand aligned, grounded responses.
- Built the backend on Supabase (PostgreSQL, real-time subscriptions) and routed model calls through OpenRouter, keeping median response time around 1.4 s.

PROJECTS

Improving NER with a Mixture-of-Experts Layer Python, PyTorch, HuggingFace | [GitHub](#) – Implemented a Mixture-of-Experts layer from scratch (MLP expert sub-networks with a learned gating network) and integrated it into a BiLSTM tagger to improve named-entity recognition.

- Sourced and prepared the training data, and ran ablation studies comparing the MoE approach against the closest comparable baselines for the task.
- Combined the MoE layer with contrastive learning to push performance beyond the standard mixture-of-experts setup.

Instruction-Tuned LLM for Domain-Specific QA Python, PyTorch, HuggingFace, FAISS | [GitHub](#) – Fine-tuned LLaMA with LoRA (parameter-efficient fine-tuning) on a curated domain-specific instruction dataset. – Added a FAISS retrieval-augmented pipeline over the domain corpus to reduce hallucination and improve factual grounding. – Evaluated with BLEU and ROUGE; LoRA + RAG outperformed zero-shot prompting on answer relevance and factual accuracy.

CO-CURRICULAR

AlgoManiax, Competitive Programming Club, BITS Goa (Core Member, Apr 2025 – Present): mentor peers in competitive programming and organize ICPC-format mock contests and team practice sessions.

TECHNICAL SKILLS

Languages: Python, C++, TypeScript, SQL, Bash

ML / AI: PyTorch, TensorFlow, HuggingFace Transformers, Fairseq, LoRA / PEFT, Scikit-learn, NumPy **LLM & NLP:** RAG pipelines, LangChain, FAISS, prompt engineering, IndicTrans2, BLEU / chrF / ROUGE evaluation **Systems &**

Web: React, Flask, REST APIs, Supabase (PostgreSQL), Git, Linux, OpenCV.